MIL-I-85080/2(AS)
28 November 1977

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, NONHEAT SHRINK, SILICONE RUBBER, FLEXIBLE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the sleeving described herein shall consist of this document and the issue in effect of MIL-I-23053.

REQUIREMENTS:

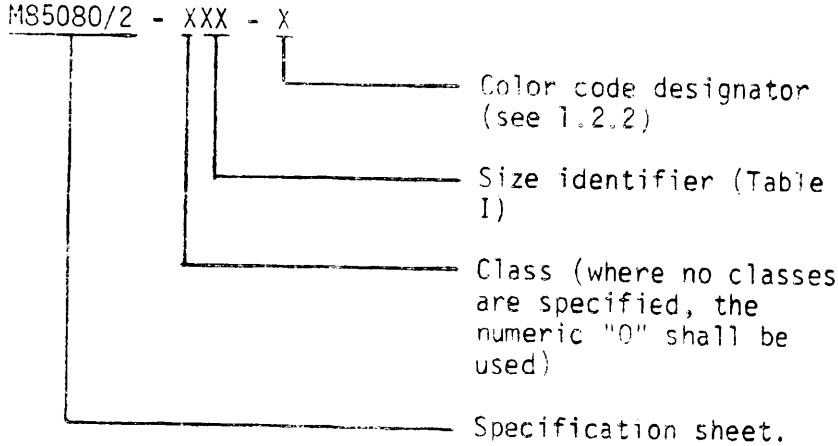
Continuous Operating Temperature range: -75°C(-103°F) to +175°C (347°F)

Mode of shrinkage: The heatless shrink tubing shall recover to a predetermined diameter upon exposure to the atmosphere. No external sources of heat are required.

Color: The tubing shall be furnished in a black color that conforms to the requirements of Class II of MIL-STD-104.

Longitudinal Change: +3%
-10%

Military part number: The military part number shall consist of the basic number of this specification sheet and dash numbers as shown below:



Example: Black, 0.27 inch as supplied I.D. sleeving shall be identified as M85080/2-005-0.

FSC 5970

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Table I. Dimensions (inches) 1/

Military Part Number 2/	As supplied I.D min 3/	After unrestricted shrinkage	
		I.D. max	Wall thickness 4/
M85080/2-001-0	.05	.031	.020 ± .010
M85080/2-002-0	.09	.062	.020 ± .010
M85080/2-003-0	.13	.093	.025 ± .010
M85080/2-004-0	.18	.125	.035 ± .010
M85080/2-005-0	.27	.187	.045 ± .010
M85080/2-006-0	.36	.250	.045 ± .010
M85080/2-007-0	.54	.375	.055 ± .010
M85080/2-008-0	.72	.500	.065 ± .010
M85080/2-009-0	.91	.625	.070 ± .010
M85080/2-010-0	1.10	.750	.090 ± .015
M85080/2-011-0	1.45	1.000	.107 ± .020
M85080/2-012-0	1.81	1.250	.110 ± .020
M85080/2-013-0	2.18	1.500	.110 ± .020
M85080/2-014-0	2.54	1.750	.110 ± .020
M85080/2-015-0	2.90	2.000	.110 ± .025

1/ Diameter limits for object to be enclosed should be in accordance with the recommendations of the manufacturer.

2/ The color code identified is the standard procurement color.

3/ The inner diameter of the sleeve may be mechanically expanded 3.0X the recovered inside diameter when in the as-supplied condition. The manufacturer should be consulted for special applications and recommendations.

4/ Wall shrinkage will be less if shrinkage is restricted.

Unrestricted shrinkage: 4.6.3.13; 23 ± 1°C(73.4 ± 2°F) for 12 hours.

TABLE II. Properties after unrestricted shrinkage

Property	Requirement	Test Conditions
Restricted shrinkage	No cracks	4.6.3.2
Voltage Withstand	Pass	4.6.6
Dielectric strength, volts per mil min.	300	4.6.2; ASTM D 876
Volume resistivity, OHMS-CM, min.	10^{11}	4.6.2; ASTM D 876
Water absorption, % max	1.0	4.6.2; ASTM D 570, Par 6.1
Fungus resistance	Non nutrient	4.6.2; ASTM G 21
Low temperature flex	No cracking	4.6.6; -75 ± 2°F (-103 ± 2°F)
Heat shock	No cracks, flowing or dripping	4.6.7; 300 ± 4°C (572 ± 8°F)

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TABLE II. (Continued)

Property	Requirement	Test conditions
Properties after heat resistance test		
Tensile strength, psi, min	450	4.6.8, 200 ± 2°C (392 ± 6°F)
Ultimate elongation, %, min	200	
Corrosion	No corrosion or pitting	4.6.9
Properties after fluid resistance test		4.6.10
Tensile strength, psi, min	300	
Ultimate elongation, %, min	100	
Odor	No objectionable odor	3.6.11
Tensile stress, @ 200% elongation, psi, max	1000	4.6.2.2 ASTM D 412
Tensile strength, psi, min	300	4.6.13 ASTM D 638 Speed D
Ultimate elongation, %, min	400	4.6.13 ASTM D 638 Speed D
Flammability	Self extinguishing; one minute max. No more than 25% of indicator flag burned.	4.6.14.1

Shelf Life (As supplied): Test method 4.6.16.

Preparing Activity
Navy - AS
(Project No. 5970-N528-2)

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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